Momen Sami Jabr

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SUMMARY

Mechatronics Engineer with professional experience in engineering standards, technical evaluation, and applied problem solving, Complete Data Science and Machine Learning through the AXSOS Academy program. Strong Python foundation for data analysis, preprocessing, model evaluation, and numerical validation using NumPy, Pandas, and Scikit-learn. Able to combine engineering logic, structured reasoning, and analytical thinking to evaluate technical problems, contribute to AI-related engineering projects, and support business intelligence, data reporting, and workflow automation using tools such as Power BI, Tableau, APIs, webhooks, CRM workflows, and low-code/no-code automation platforms.

TECHNICAL SKILLS

Languages: Python, SQL.

Databases: Microsoft SQL Server

Python Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, imbalanced-learn, TensorFlow/Keras

Engineering & Technical Background: Mechatronics Engineering, Electrical Systems Fundamentals, Digital Logic, Circuits, PLC Fundamentals, Technical Standards Review

Tools: Jupyter Notebook.

BI & Automation Tools: Power BI, Tableau, n8n, CRM Automation, Webhooks, API-based Integrations, Low-Code/No-Code Workflow Design

Core Strengths: Core Strengths: Data Analysis, Machine Learning, Problem Solving, Technical Evaluation, Numerical Validation, Data Visualization, Workflow Automation, Business Intelligence Reporting

Languages: Arabic (Native), English (B2/C1)

TECHNICAL PROJECTS

Prediction of Product Sales | Data Scientist | Link

A machine learning project focused on analyzing historical retail data and predicting future product sales.

- Built a regression workflow to analyze historical retail data and predict Item_Outlet_Sales using Python and Scikit-learn.
- Performed data cleaning, feature preparation, and exploratory data analysis to identify sales patterns, skewness, and relationships between product/outlet variables and the target.
- Developed end-to-end preprocessing and modeling pipelines using imputers, scaling, encoding, and ColumnTransformer to prevent data leakage.
- Trained and compared Linear Regression and Random Forest Regressor models, then optimized Random Forest with GridSearchCV and evaluated performance using MAE, RMSE, and R^2.

Metabolic Syndrome Prediction | Data Scientist | Link

A machine learning project focused on predicting metabolic syndrome risk from clinical and demographic health data.

- Built and evaluated classification models to predict metabolic syndrome from NHANES-derived clinical and demographic data using Python and Scikit-learn.
- Performed preprocessing and exploratory analysis on mixed numerical and categorical health features, including imputation, scaling, encoding, class-distribution analysis, and correlation analysis.
- Compared multiple models including Random Forest, Logistic Regression, and Artificial Neural Networks, with particular attention to positive-class recall due to the medical screening nature of the task.
- Used metrics such as precision, recall, F1-score, accuracy, and confusion matrices to compare model trade-offs, and applied SMOTE and Keras Tuner in ANN experiments.

PROFESSIONAL EXPERIENCE

Palestinian Standards Institution | *Mechatronics Engineer* | *Ramallah, Palestine* | 2022 – Present

- Reviewed and contributed to engineering-related standards and technical specifications in areas related to machinery, vehicles, fire safety, and fuel equipment.
- Participated in technical evaluation, standards development, and structured review processes requiring engineering judgment, accuracy, and compliance-oriented thinking.
- Worked on multidisciplinary engineering topics involving mechanical, electrical, and safety considerations, strengthening technical analysis and professional problem-solving skills.
- Supported technical committees and engineering documentation activities with a focus on clarity, consistency, and applied engineering logic.

EDUCATION

Axsos Academy | *Data Science & Machine Learning Program*| *Ramallah, Palestine* 2025-2026

- *A part-time immersive training program focused on data analysis, statistics, and machine learning using Python. The bootcamp includes 20 weeks of hands-on training covering +620 hours in total, including data manipulation and analysis with Pandas and NumPy, exploratory data analysis (EDA), data visualization, probability and statistics, SQL using Microsoft SQL Server, and applied machine learning. The program includes individual and group-based projects, real-world datasets, and extensive practice in problem-solving, model evaluation, and data-driven decision making, alongside continuous technical assessments and coding exercises.*

An-Najah National University | *Bachelor in Mechatronics Engineering* | *Nablus, Palestine* 2014-2019

- *A five-year bachelor's program combining mechanical, electrical, control, and computer-based engineering concepts. Coursework included mechatronics systems, sensors and actuators, control systems, circuits, digital logic, PLC fundamentals, technical drawing, mathematics, and practical engineering problem-solving, with emphasis on integrating hardware and software for automated and electromechanical systems.*

- *Relevant Coursework: C Programming, MATLAB, Digital Logic, Electrical Circuits, PLC Fundamentals, Control Systems, Assembly Language*